Classical Field Theory — Exercise Sheet 20

Nonabelian constraints and local reduction

Hints and answers

Use $T_a = -\mathrm{i}\sigma_a/2$ for $\mathfrak{su}(2)$, so $[T_a, T_b] = \epsilon_{abc}T_c$ and $\langle T_a, T_b\rangle = \delta_{ab}$. All spatial fields are smooth on $\mathbb{T}^3$. Write $\mathcal{V}_{\mathrm{per}} = (2\pi)^3$ for its volume. The canonical bracket has the order fixed in Lectures 7 and 20.

Exercise 20.1 (A homogeneous constraint manifold). Restrict the spatial fields to constants $A_i = a_i^a T_a$ and $\Pi_i = p_i^a T_a$. A parameter $\varepsilon = \varepsilon^a T_a$ is also constant.

(a) Pull the canonical symplectic form back to these constants and derive their finite-dimensional bracket. Compute the generator $C[\varepsilon] = \mathcal{V}_{\mathrm{per}} \sum_i \langle \Pi_i, [A_i, \varepsilon]\rangle$ and its Hamiltonian vector field. Prove its constraint bracket, including the volume factor and sign.

(b) At $A_i = T_i$ for $i = 1, 2, 3$, write $\Pi_i = \sum_j M_{ij}T_j$ for a real matrix M. Classify all M satisfying $\sum_i [A_i, \Pi_i] = 0$. Give one nonscalar matrix in the constraint set and one outside it. Explain why checking a single homogeneous state cannot prove the constraint algebra on the full periodic field space.

Hint. A homogeneous field variation is constant, so its spatial integral contributes the volume to the symplectic form. Pair the two terms involving M_{ij} and M_{ji} in each Gauss component.

Answer. The bracket is $\mathcal{V}_{\mathrm{per}}^{-1} \sum_{i,a} (F_{a_i^a} G_{p_i^a} - F_{p_i^a} G_{a_i^a})$. The generator gives $\delta A_i = [A_i, \varepsilon]$ and $\delta\Pi_i = [\Pi_i, \varepsilon]$, with $\{C[\varepsilon], C[\zeta]\} = -C[[\varepsilon, \zeta]]$. At $A_i = T_i$, Gauss holds exactly when M is symmetric.

Exercise 20.2 (Integrating a residual gauge generator). For a fixed smooth periodic $\varepsilon(x)$, let $h_s(x) = \exp(s\varepsilon(x))$ and define

$$A_i(s) = h_s^{-1} A_i h_s + h_s^{-1}\partial_i h_s, \qquad \Pi_i(s) = h_s^{-1}\Pi_i h_s.$$

Here s is the transformation parameter, not physical time.

(a) Prove that these arrays solve $\partial_s A_i(s) = D_i^{(s)}\varepsilon$ and $\partial_s \Pi_i(s) = [\Pi_i(s), \varepsilon]$. Show that the transformation preserves the canonical form, the temporal-gauge Hamiltonian and the Gauss constraint surface.

(b) Take $\varepsilon = \sin x^1\, T_3$, $A_1 = T_1$, $A_2 = T_3$, $A_3 = 0$, $\Pi_1 = T_2$ and $\Pi_2 = \Pi_3 = 0$. Compute the transformed fields and their Gauss array. Is this example on the constraint surface? What changes if a finite gauge transformation depends on physical time?

Hint. Differentiate $h_s^{-1}h_s = 1$ and use $h_s\varepsilon = \varepsilon h_s$. For the constraint use $D_i^{(s)}(h_s^{-1}Xh_s) = h_s^{-1}(D_iX)h_s$. Check the derivative part of the transformed Gauss array as well as its commutator.

Answer. The flow is generated by $C[\varepsilon]$ and preserves the stated structures. With $\alpha = s\sin x^1$,

$$A_1(s) = \cos\alpha\, T_1 - \sin\alpha\, T_2 + s\cos x^1\, T_3, \quad A_2(s) = T_3,$$
$$\Pi_1(s) = \sin\alpha\, T_1 + \cos\alpha\, T_2, \qquad \sum_i D_i^{(s)}\Pi_i(s) = T_3.$$

The example violates Gauss. Temporal gauge is preserved by a finite transformation exactly when its physical-time derivative is zero.

Exercise 20.3 (A linearized reduced bracket and its limit). Fix a real constant η and the background $A_1^0 = \eta T_3$, $A_2^0 = A_3^0 = 0$, $\Pi^0 = 0$. Work only with smooth variations a_i, p_i and parameters having zero spatial mean. This excludes the entire zero Fourier mode from this problem. Put $D_i^0 = \partial_i + [A_i^0, \cdot\,]$.

(a) Derive the linearized Gauss equation $D_i^0 p_i = 0$ and the gauge variation $a_i \mapsto a_i + D_i^0 \varepsilon$. For the Coulomb condition $\partial_i a_i = 0$, compute the eigenvalues of $M = -\partial_i D_i^0$ on each nonzero Fourier mode, using $T_+ = T_1 + \mathrm{i}T_2$, $T_- = T_1 - \mathrm{i}T_2$ and T_3.

(b) For $|\eta| < 1$, prove that M^{-1} maps smooth zero-mean arrays to themselves. Define

$$P = 1 + D^0 M^{-1}\,\mathrm{div}, \qquad P^* = 1 + \nabla M^{-1} D^0 \cdot\,.$$

Prove that Pa is divergence-free and gauge invariant, while $P^* p$ satisfies linearized Gauss. Prove $P^2 = P$. For real smooth zero-mean vector smearings f, g, compute the canonical bracket of $\mathcal{A}[f] = \int \langle f, Pa\rangle$ and $\mathcal{P}[g] = \int \langle g, P^* p\rangle$.

(c) At $\eta = 2$, find a real zero-mean parameter with $M\varepsilon = 0$ but $D^0\varepsilon \neq 0$. Explain why this prevents using the preceding inverse there. Does the calculation establish a global nonlinear reduced bracket?

Hint. The adjoint eigenvalues of T_3 on T_+ and T_- are $-\mathrm{i}$ and i. Use periodic integration by parts for P^*. For the last part try a parameter depending on $x^1 + x^2$.

Answer. At $n \in \mathbb{Z}^3 \setminus \{0\}$, the three eigenvalues are $|n|^2 - \eta n_1$, $|n|^2 + \eta n_1$ and $|n|^2$. For $|\eta| < 1$,

$$\{\mathcal{A}[f], \mathcal{P}[g]\} = \int_{\mathbb{T}^3} \langle f, Pg\rangle \,\mathrm{d}^3x.$$

At $\eta = 2$ one choice is $\varepsilon = T_1 \cos(x^1 + x^2) - T_2 \sin(x^1 + x^2)$. The result is a linear reduction in the specified Fourier sector, not a global nonlinear gauge construction.